Cheap: measured on the doser itself

Physics-based ML

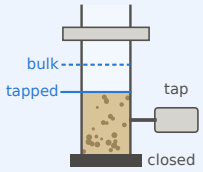
surrogate models and DEM simulations find the links

Feed factor

mg per auger revolution at several tilts, and before/after humid air

measured now (13 powders)

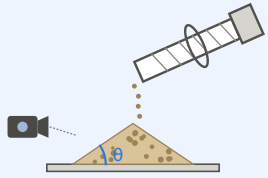
"Ghost dose"



transparent doser, outlet closed, tapped: bulk and tapped density

proposed: thin-walled SLA vessel

Angle of repose



camera on the pile the doser pours

proposed: camera at the outlet

State and history of the powder

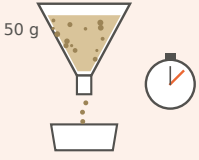


composition; how it was made, sieved, dried, stored or reused

infer the properties of a new powder

known properties speed up calibration

Hall/Carney flow



50 g through a standard funnel (ASTM B213, B964)

in-house: funnel set

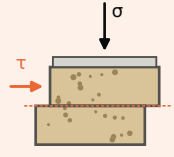
Particle size (PSD)



SEM images (BSE mode) or laser diffraction

in-house SEM, or outside lab

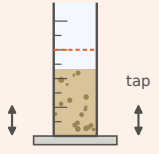
Shear cell, ff_c



cohesion under load (ASTM D6773, D7891)

outside lab

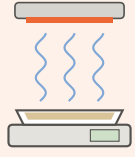
Tapped density



graduated cylinder, loose then tapped (ASTM B527)

in-house: cylinder

Moisture



loss on drying (weigh, dry, reweigh)

in-house: oven, balance

Standard tests: a separate instrument for each, some only in outside labs